

ME 207 | Fluid Dynamics

Computational Fluid Dynamics Project

Due: Midnight of April 17, 2025

Instructions

- This is an individual project. Each student should upload the report and the associated Ansys files in any cloud storage (Google Drive or Microsoft One drive) and share the link by submitting this Google form (Link to Google Form will be shared later).
- The report must consist of (1) geometry and boundary conditions, (2) mesh statistics, (3) discretisation schemes and solution methodology, (4) results and discussion, and (5) conclusion.
- All the figures must be included in the results and discussion section (except mesh independence), and adequate physics-based reasoning should be provided.
- Report must be prepared using WORD or LaTeX. Handwritten reports will not be accepted.

Problem Statement

Consider laminar flow over a NACA 0012 airfoil section. The chord length of the airfoil is 1 meter. The freestream velocity is 25 m/s and the air temperature is 20°C.

1. Create a 2D computational domain including the NACA 0012 airfoil section at 0° angle of attack. Choose a sufficiently fine mesh for the discretization of the domain. Apply appropriate boundary conditions (inlet, outlet, and symmetry or wall boundaries).
2. Calculate drag coefficient (C_D) and lift coefficient (C_L) for 0° angle of attack (AOA).
3. Simulate for 5° AOA and calculate drag and lift coefficients (you can achieve this by modifying the inlet velocity components based on AOA)
4. Tabulate your results for 0° and 5° AOA and compare your results with the literature as shown below,

AOA	Parameters	Ansys – CFD results	Literature
0°	C_D		
	C_L		
5°	C_D		
	C_L		

5. Plot the velocity, pressure and streamline contours separately for each case.
6. Mark and describe the flow features in the contours (including boundary layer and flow separation if any).
7. Using the insights obtained from CFD simulations, explain the observed variations of lift and draft coefficients with angle of attack.